

Received Signal Modeling and BER Analysis for Molecular SISO Communications

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ABSTRACT

A well-modeled molecular communication (MC) channel is crucial for the development of required equipment. Previous models, however, have often considered ideal receivers with perfect absorption, which resulted in impractical results. Here, we propose an improved model consisting of a partially absorbing receiver with four unique parameters, whose values are determined using Particle Swarm Optimization (PSO). We evaluate the Root Mean Square Error (RMSE) performance of our model in the estimated received molecules, which shows consistent results for different transmitter-to-receiver distances or receiver radii, besides a five times improvement in the cumulative number of received molecules compared to that of previous models. We also consider two new Concentration Shift Keying (CSK) modulation schemes, in addition to typical ones, and analyze the trend of Bit Error Rate (BER) and detection threshold for these schemes at different channel lengths. Finally, we investigate the feasibility of Hamming code correction on these schemes and show that the third-order Hamming-corrected BCSK (Binary CSK) modulation is the best for the proposed SISO model.

CCS CONCEPTS

• **Networks** → **Network performance analysis**.

KEYWORDS

Molecular Communication, Partially Absorbing Receiver, Concentration Shift Keying, Particle Swarm Optimization, Hamming Code Correction.

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1 INTRODUCTION

For electromagnetic (EM) communication, the wavelength is proportional to antenna size, which is hard to afford at the micro/nanoscale [6]. Hence, molecular communication (MC) has received significant attention at this scale thanks to offering miniature size, low energy consumption, and high biological compatibility [1], thereby enabling various applications, e.g., in-vivo drug delivery and pipeline leakage monitoring [9].

As the name reveals, MC uses molecules to transmit, receive, and decode information, where diffusion is the easiest way of data transfer [4]. Due to the stochastic nature of Brownian motion, however, diffusion-based MC suffer from inter-symbol interference (ISI), which is caused by a low signaling interval, i.e., a new signal is transmitted before the previously sent signal is processed. If the ISI is not considered, the MC system performance may severely degrade. Hence, modeling the number of received molecules, especially under changing channel conditions, is crucial to eliminate the ISI effect [3, 10, 13].

Recently, machine learning (ML)-based methods have gained notable momentum in this area due to their promising performance in the case of no or limited knowledge of the underlying channel. In [14], an ML-based molecular 2×2 multiple-input multiple-output (MIMO) channel model with a point transmitter and a perfectly absorbing spherical receiver was proposed. Despite the potential, this model had some limitations, i.e., the perfectly absorbing receiver assumption was impractical. For the same transmitter-receiver pair, [19] showed that adding compensation parameters could achieve better results in terms of the Root Mean Square Error (RMSE) of the received molecules. Similarly, [16] proposed a data-driven, model-based receiver, which was again assumed to be perfectly absorbing.

Although the results showed that their design could perform similar to the receivers with prior information about the underlying channel, it was impractical due to the same assumption. The first study tackling this drawback was [2], which proposed a partially absorbing spherical receiver taking all the hitting molecules in but counting only the ones that hit from a specific angle.

In addition to the studies focusing on receiver optimization, considerable efforts have been made toward enabling different modulation schemes in MC. One key method is Concentration Shift Keying (CSK), which has two common types: Binary CSK (BCSK) and Quadrature CSK (QCSK) [12].

In [14], the Bit Error Rate (BER) of a BCSK modulation was calculated in a MIMO channel. When the same scheme was used for a single-input single-output (SISO) setting, the BER turned out to be high; thus, the research has looked for complementary methods to lower the BER. One of them was Hamming code correction, studied in [15], which proposed a method to enhance the performance of a diffusion-based MC system through higher-order Hamming codes but did not consider the effect of changing channel length, which was kept constant.

Considering these, this paper proposes a diffusion-based molecular SISO channel with a partially absorbing spherical receiver and transmitter (Sec. 2.1). Our model is inspired by [2], but we aim for a better estimation of the number of received molecules, which is validated using the dataset from [19]. In our model, we introduce a receiver-specific parameter called “angle of reception”, besides some additional parameters that act as compensation parameters. We use Particle Swarm Optimization (PSO), a fast and effective technique to search for the optimal values of these parameters (Sec. 2.2), to prove that our model can predict the number of received molecules with a lower RMSE for the envisioned channel model (Sec. 3.2). For such a SISO channel, we report the results on the typical CSK modulation schemes (BCSK and QCSK) in literature as well as the ones that we propose, Hexal CSK (HCSK) and Octal CSK (OCSK), which answer whether the higher order modulation is more advantageous or not. Those allow us to verify the trend of minimum BER for various channel lengths and corresponding one-bit detection thresholds for the CSK schemes (Sec. 3.3). We also compare those schemes in terms of Hamming code correction, where the results reveal using third-order Hamming codes with BCSK improves the BER to a practically usable value (Sec. 3.4). Lastly, we provide our conclusions and identify future research directions (Sec. 4).

2 SYSTEM MODEL

This section describes our proposed SISO channel model and explains how we use the PSO learning algorithm to obtain the channel parameters required.

2.1 Channel Model

As shown in Fig. 1, our model consist of a spherical receiver (Rx) and a transmitter (Tx), where the diffusion coefficient of the medium between them is D . Unlike the existing studies in literature, such as [19] and [5], we do not assume a perfectly absorbing receiver in our model. Instead, our receiver is modeled to be partially absorbing at an angle of reception, β . Also, we assume that the transmitter is

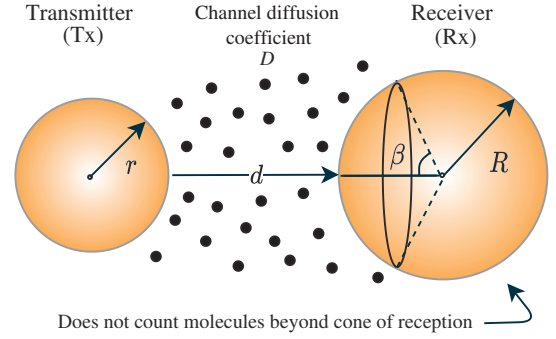


Figure 1: Molecular SISO channel with spherical Tx and Rx.

perfectly aligned with the receiver at a distance of d to each other, and the radii of Tx (r) and Rx (R) vary from 5 to $10\mu\text{m}$.

For the ideal case assuming full absorption, the fraction of molecules that hit the receiver till time t can be given as [19]:

$$F(t) = \frac{R}{d+R} \times \text{erfc}\left(\frac{d}{y}\right), \quad (1)$$

where $\text{erfc}(\cdot)$ is the complementary error function, and y is equal to $\sqrt{4Dt}$. However, this model returns inaccurate results, i.e., high RMSE, for small distances between Tx and Rx. Thus, to achieve a more accurate model, (1) should be modified as incorporating a cone of reception with angle β , where the receiver's contribution outside this cone is negligible. Hence, in this case, the cumulative number of received molecules for this receiver can be written as [2]:

$$F(\beta, \tau) = \frac{R}{d+R} \times \text{erfc}\left(\frac{d}{y}\right) \times \frac{\Omega(\tau, y, \beta)}{U(\tau, y)}, \quad (2)$$

where Ω is:

$$\begin{aligned} \Omega(\tau, y, \beta) = & \frac{1}{d} \text{erfc}\left(\frac{d}{y}\right) - \frac{1}{x(\beta)} \text{erfc}\left(\frac{x(\beta)}{y}\right) \\ & + \frac{1}{\sqrt{2\pi D\tau}} \left[\text{Ei}\left(-\frac{d^2}{y^2}\right) - \text{Ei}\left(-\frac{x^2}{y^2}\right) \right], \end{aligned} \quad (3)$$

with

$$x(\beta) = \sqrt{(R+d)^2 - 2(R+d)R \cos(\beta) + R^2},$$

and

$$\begin{aligned} U(\tau, y) = & \frac{1}{d} \text{erfc}\left(\frac{d}{y}\right) - \frac{1}{d+2R} \text{erfc}\left(\frac{d+2R}{y}\right) \\ & + \frac{1}{\sqrt{D\tau}} \left[\text{Ei}\left(-\frac{d^2}{y^2}\right) - \text{Ei}\left(-\frac{(d+2R)^2}{y^2}\right) \right], \end{aligned} \quad (4)$$

in which $\text{Ei}(\cdot)$ is the exponential integral function.

To enhance the channel model given in (2) for a more practical case, we introduce three compensation parameters, namely overall scaling b_1 (called also model fitting), and $4D$ to the power of b_2 and τ to the power of b_3 in y , where τ is time. Thus, (2) can be reexpressed including the newly introduced b_i parameters as:

$$F'(\beta, b_i, \tau) = b_1 \times \frac{R}{d+R} \times \text{erfc}\left(\frac{d}{y'}\right) \times \frac{\Omega(\tau, y', \beta)}{U(\tau, y')}, \quad (5)$$

where $y' = (4D)^{b_2} \tau^{b_3}$. All in all, our model has a total of four parameters, which are β , b_1 , b_2 , and b_3 .

2.2 Learning Problem

We use Least Square Error (LSE) loss function to minimize the error between the model output and the simulation data in regards to the parameters β , b_1 , b_2 , and b_3 . Hence, our learning problem can be given as:

$$\sum_{k=1}^M (F'(\beta, b_i, \tau_k) - S(\tau_k))^2, \quad (6)$$

where $S(\tau)$ is the mean simulation data representing the ratio of molecules hitting the receiver till time τ . The output of the learning process consists of model parameters to be used in the following sections. Note that (6) cannot be used as a loss function in the standard ML procedures since (5) is a non-smooth function.

Thus, we use PSO, which does not rely on gradient computations or any differentiation of the objective function and has fewer hyperparameters to optimize [8]. Hence, PSO can find the best channel model parameters that minimize (6) for a given dataset by searching the parameter space.

3 CHANNEL ESTIMATION AND BER CALCULATIONS

In this section, we give a detailed description of the dataset and the PSO hyperparameters used during our search and discuss the outcomes of our evaluation. We also perform the BER analysis for different CSK modulation schemes and investigate the order of Hamming code correction that is best suited for our given channel model.

3.1 Dataset and parameters used

The dataset being used in this paper is sourced from [19], which contains the number of simulated molecules in the receiver for a SISO channel with a spherical Tx-Rx pair. The number of received molecules is plotted in time steps of ΔT for various values of d , R , and D , as listed in Table 1. Here, we have three common parameters, i.e., b_1 , b_2 , and b_3 , which do not depend on input parameters d , R , and D . Different from [19], we have an additional parameter (β) depending on d , R , and D , which hinders parameter training based on the entire dataset since this process requires all parameters to

Table 1: Parameter values.

Parameter	Value
No of Transmitted Molecules	3000
Simulation time step, ΔT	50 μs
Distance between Tx and Rx, d	2, 4, 6, 8, 10 μm
Rx radii, R	5, 7.5, 10 μm
Tx radii, r	5, 7.5, 10 μm
Diffusion coefficient, D	50, 75, 100 $\mu m^2/s$
PSO cognitive parameter	0.5
PSO social parameter	0.3
PSO inertia parameter	0.9

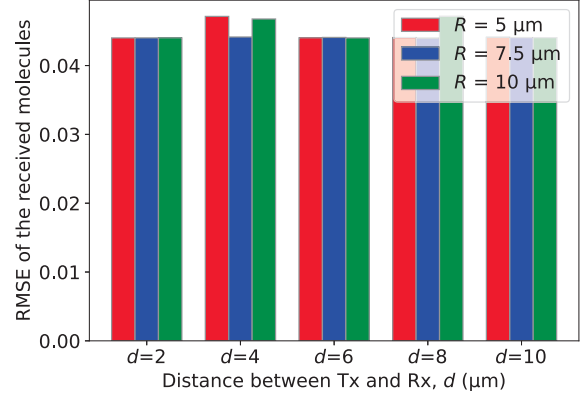


Figure 2: RMSE in the estimated number of received molecules vs. d for different R values.

be common, i.e., independent of input parameters. Thus, we train the model for different cases of d , R , and D using the dataset from [19]. Here, the parameters b_1 , b_2 , and b_3 will be common for the entire dataset, while β will vary from case to case in the range of 0° - 90° . The MSE is calculated for our proposed model and the one in [19] by searching the parameter space and finding the best fit concerning the loss function in (6). We run 200 iterations for each case of the PSO [8] using the PSO hyperparameters and the dataset parameters listed in Table 1.

3.2 Channel model evaluation

Fig. 2 plots the RMSE of our model for different values of d and R at $D = 50 \mu m^2/s$ ¹. The RMSE is found by taking the square root of the MSE derived from (6) based on the optimal parameters obtained from the PSO search.

As seen, the error is close to 0.44 for almost every case, which proves that our model is consistent over d and R , i.e., it is not affected by Tx-to-Rx distance or Rx radius. That is a major improvement compared to [19] since their RMSE depends on both d and R , giving high values especially when $d < 4 \mu m$. Hence, our model returned five times lower RMSE in the estimated cumulative number of received molecules compared to that of [19]. This improvement is clearly visible from Fig. 3, which shows that the cumulative number of received molecules diverges by a large factor in Yilmaz et al.'s estimation [19] from the simulation data (at $d = 2 \mu m$, $R = 5 \mu m$, and $D = 50 \mu m^2/s$). We also see that the model in [2] performs worse than both our model and the one in [19] as it does not have the compensation parameters.

Our experiments reveal that b_1 , b_2 , and b_3 are nearly constant for different values of d , R , and D . However, β may vary from case to case². Specifically, it does not change much with D but reduces with increasing R , which is intuitive. Note that we do not initially constrain the simulation to a specific β as in [2]. Instead, we model the generic spherical absorbing receiver into a partially

¹A specific D ($50 \mu m^2/s$) is chosen since the RMSE does not significantly vary with varying D [19].

²The optimum parameter values can be found at <https://doi.org/10.5281/zenodo.7036057>.

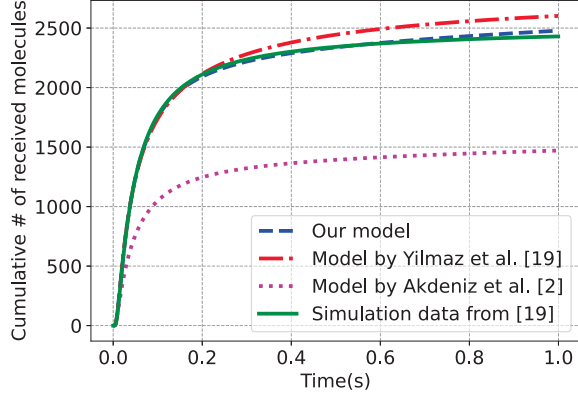


Figure 3: Comparison of our work with [19], [2], and simulation data in terms of the cumulative number of received molecules.

absorbing one, which yields the optimal value of β through PSO. Next, we investigate the BER of our channel for different CSK modulation schemes, which is crucial for designing a reliable MC channel model.

3.3 BER calculations

In [11], the authors considered the typical CSK modulation schemes, i.e., BCSK and QCSK, for BER analysis. Here, we propose two new schemes, namely HCSK and OCSK, inspired by [17], to understand the trend of minimum BER vs. optimal channel length and the corresponding detection thresholds for different CSK modulation schemes.

Our definition of CSK is similar to that of [5]. Specifically, a fixed range of the number of received molecules translates into one possible transmitted bit. Hence, for a detection threshold of ξ received molecules, the k -CSK scheme ($k = 2, 4, 6, 8$) is defined as follows for N received molecules:

$$\begin{aligned} N < \xi &\Rightarrow 0 \\ \xi \leq N < 2\xi &\Rightarrow 1 \\ 2\xi \leq N < 3\xi &\Rightarrow 2 \\ &\vdots \\ (k-1)\xi \leq N &\Rightarrow k-1 \end{aligned}$$

Thus, in BCSK, we need only one threshold to detect bits between 0-1. In QCSK, we need three thresholds to detect symbols in the range 0-3, which are five and seven for HCSK and OCSK for the ranges of 0-5 and 0-7, respectively.

In an MC channel of fixed length, the BER of transmission is minimum for a specific detection threshold, which is the optimum operating case. The channel length determines the frequency of transmission, and the detection threshold is a technological limitation depending on the receiver's accuracy. To benchmark these, our first goal is to compare the minimum BER values concerning channel length for different CSK modulation schemes and determine the detection thresholds at those minimum values.

3.3.1 BER from data. From [14] (originally done for a 2×2 MIMO channel), we calculate the BER for a SISO channel with ISI and

then extend this idea to different modulation schemes. With $\Delta T = 5 \times 10^{-4}$ sec, the symbol slots are of length $n\Delta T$ with channel length n having different integer values. The channel length is related to the frequency of transmission. For a given ISI window, the number of received molecules at the m^{th} symbol slot is the sum of Binomial distributed random variables, where each summand represents the number of hitting molecules due to a previous emission and thus accounts for the ISI.

Here, we consider an ISI window of η . For a large enough m , i.e., the total number of symbol slots, we can use Gaussian approximation on a Binomial distribution. In [18], a similar calculation was done assuming Poisson distribution. This study showed that there was not much difference between a Poisson process and Gaussian approximation (Fig 10. in [18]) except for a slight shift in the threshold. Also, the BER vs SNR curves turned out to be remarkably similar. Thus, the general case can be written as:

$$N^{Rx}[m] \sim \text{Normal}(\mu, \sigma^2),$$

where

$$\mu = \sum_{k=0}^{\eta} N s_{m-k}^{Tx} F[k], \quad (7)$$

$$\sigma^2 = \sum_{k=0}^{\eta} N s_{m-k}^{Tx} F[k](1 - F[k]), \quad (8)$$

From this, the BER (P_e) can be given as:

$$P_e = \frac{1}{n} \times \sum_{s_{m-\eta:m}^{Tx}} P_{e|s_{m-\eta:m}^{Tx}}, \quad (9)$$

in which

$$P_{e|s_{m-\eta:m}^{Tx}} = P(s_m^{Tx} \neq s_m^{Rx} | s_{m-\eta:m}^{Tx}), \quad (10)$$

where s_m^{Tx} is the transmitted and s_m^{Rx} is the received bit corresponding to the m^{th} symbol slot. For our calculation purposes, we take η as 5.

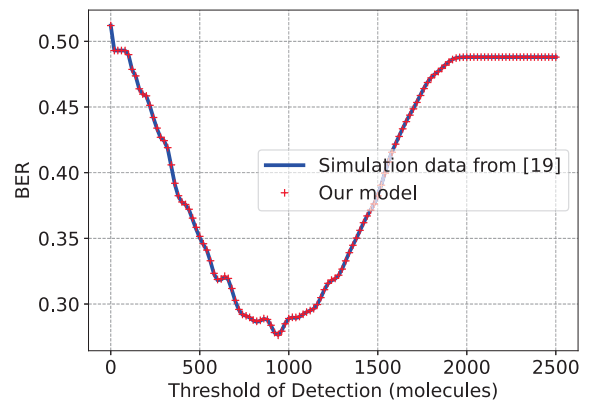


Figure 4: BER of the BCSK vs. detection threshold with a channel length of 40 and ISI window of 5.

3.3.2 BER of a specific case. Fig. 4 compares our model with the simulation data from [19] in terms of BER of the BCSK modulation for an increasing molecule detection threshold. Here, the parameters are selected as $d = 2\mu\text{m}$, $r = R = 5\mu\text{m}$, and $D = 50\mu\text{m}^2/\text{s}$, which returned the maximum RMSE in [19], while in our model, the RMSE is independent of these parameters, as shown in Fig. 2.

Fig. 4 illustrates that the BER of our model closely matches the simulated BER, verifying the low RMSE at $d = 2\mu\text{m}$ for our model. Thus, for convenience, we use the simulation data for BER estimation instead of our trained model as they practically give the same result. We also notice that there is a minimum at a specific detection threshold for the BCSK (at around 1000 molecules in Fig 4), which is the optimal operation point for this model. Next, we investigate this further for different CSK modulation schemes.

3.3.3 Minimum BER for different channel lengths. In Fig. 5, we show the minimum BER achieved by different CSK modulation schemes for increasing channel lengths with optimum detection threshold at $d = 2\mu\text{m}$, $r = R = 5\mu\text{m}$, and $D = 50\mu\text{m}^2/\text{s}$. As observed, the BER decreases almost linearly with the increasing channel length. That is primarily due to the less ISI occurring at higher channel lengths besides the distinguishability of the peaks in the incoming molecules count [7]. More importantly, the BER of a very low channel length for the BCSK is almost the same as the BER of a very high channel length for the QCSK. As a higher channel length implies a slower communication rate, and our focus is always on faster communication, BCSK seems to be the only suitable option among the four CSK modulation schemes considered. Thus, a channel length of about 80 will be sufficient to achieve low BER with the BCSK modulation.

Fig. 6 shows the optimum detection threshold vs channel length corresponding to the minimum BER values in Fig. 5. An almost linear increasing trend is observed for the detection threshold; however, this increase is tiny for higher-order CSK schemes. On the other hand, BCSK shows a considerable variation, implying that the BER is sensitive to the detection threshold. From a practical aspect, it is difficult to accurately count the number of molecules that are incident on Rx, and at best, we can give a value within

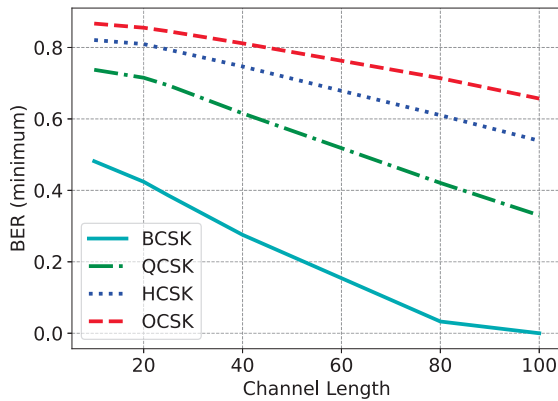


Figure 5: Minimum BER achieved vs. channel length for different CSK schemes.

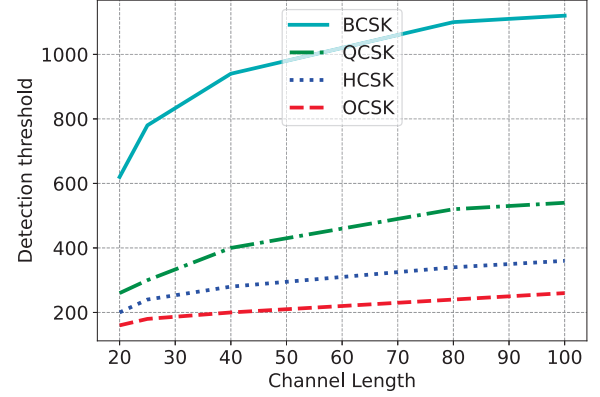


Figure 6: Optimum detection threshold vs. channel length for different CSK schemes.

an error bound. For a variation in channel length, the detection threshold of BCSK changes significantly, while OCSK remains relatively constant. This implies that for an MC with varying frequency, higher-order CSK schemes outperform the BCSK.

3.4 Hamming code correction

Inspired by [15], we implement Hamming code correction to reduce the BER of the BCSK modulation. The one-bit coded error probability after Hamming code correction is given by:

$$P_{\text{error-coded}} = \frac{1}{n} \sum_{j=t+1}^n j \binom{n}{j} p^j (1-p)^{n-j}, \quad (11)$$

where $n = 2^m - 1$ is the block length with m parity check bits ($m \geq 2$), $t = (d_{\min} - 1)/2$ is the number of errors that can be corrected, and $d_{\min} (= 3)$ is the minimum distance.

In Fig. 7, we plot the BER achieved by the four CSK modulation schemes concerning different detection thresholds, besides a fifth-order Hamming code-corrected BCSK case. It should be noted that we did not consider Hamming code correction for higher-order CSK

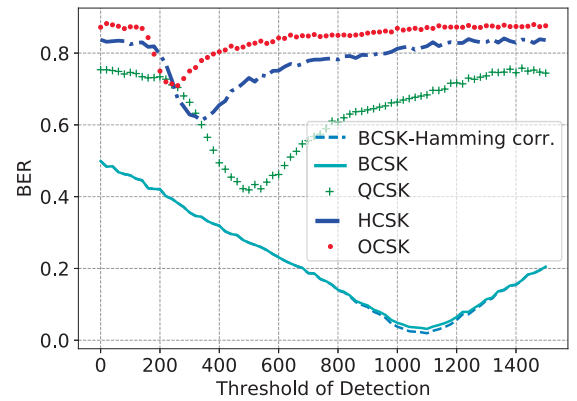


Figure 7: BER vs. detection threshold for different CSK schemes at channel length of 80.

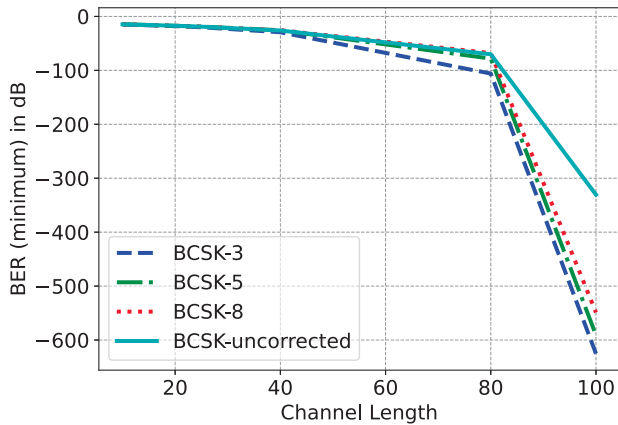


Figure 8: Minimum BER achieved vs. channel length for different orders of Hamming code correction to the BCSK.

schemes since they already have high BER, which is not improvable by Hamming codes. From Fig. 7, we observe that Hamming code correction lowers the BER of the BCSK modulation. This result compels us to answer the question “Which order of Hamming code correction is the best in our case?”. Therefore, we plot the minimum BER (in dB) for different orders of corrections to the BCSK in Fig. 8. Clearly from the figure, the third-order Hamming code correction is the most effective for the envisioned molecular SISO channel model.

4 DISCUSSIONS, CONCLUSIONS, AND FUTURE WORKS

We hereby conclude that our correction to the model proposed in [2] gave a more reliable result to estimate the diffusive molecular channel with a spherical Tx-Rx pair, with Rx being partially absorbing. We also achieved a significant improvement in the RMSE of the estimated number of received molecules, which was independent of d and R , compared to that of [19]. Furthermore, we investigated the minimum BER and the corresponding detection threshold for different CSK modulation schemes at a specific channel length. Our results suggested that the minimum BER has a linear behavior with increasing channel length for all modulation schemes. We also showed that while the BCSK outperforms other schemes in terms of channel length and BER, it has the limitations of a very high and changing detection threshold and low information capacity. These make the BCSK slow and somewhat unreliable, especially in the short distance and varying frequency regime. Based on this, we analyzed the impact of Hamming codes in reducing the BER of the BCSK and observed that a third-order Hamming-corrected BCSK performs much better at a channel length of 80. Thus, our work resolved the long-lasting channel estimation problem to a great extent and revealed the practical applicability of uncorrected and corrected CSK modulation schemes to achieve lower BER values in diffusion-based modular SISO channels.

We plan to extend this model to MIMO channels, where ML-based decoders can be used to implement the CSK modulation schemes achieving low BER values.

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